Contents

1 Percentages **2**

2 Geometry **2**
2.1 Angles in polygons . . . 2
2.2 The pythagorean theorem . . . 3
2.3 Pi . . . 4
2.4 More on angles . . . 5

3 Prime factorization **5**

4 Sets **5**

5 More algebra **6**
5.1 Simultaneous equations . . . 6
5.2 Inequalities . . . 6
5.3 Graphs . . . 7
5.4 Quadratic equations . . . 7

6 Probability and statistics **7**

7 Trigonometry and more geometry **8**
7.1 Trigonometry . . . 8
7.2 More geometry stuff . . . 9

8 Square roots **10**

9 Functions **10**
9.1 Definition . . . 10
9.2 Graphs . . . 11

10 Things to be careful of **13**

11 Sequences **14**

12 Infinity **15**

13 More on functions **15**
13.1 Inverse functions . . . 15
13.2 Composite functions . . . 15

14 Powers and roots **16**

15 Lower and upper bounds from rounding **16**

16 The quadratic formula **16**

17 Quadratic graphs **18**

18 Series **19**

19 More geometry **21**
19.1 Sine rule . . . 21
19.2 Cosine rule . . . 22
19.3 Scaling shapes . . . 22
19.4 Formulae for volumes and surface areas . . . 23
19.5 Circle properties . . . 24

20 Fractions and decimal expansions **28**

21 More algebra **28**
21.1 Difference of squares rule 28
21.2 Algebraic fractions 29
21.3 Simultaneous equations 29

22 Vectors **29**

23 Slopes and perpendicular lines on graphs **30**

1 Percentages

We can do reverse percentages. Be careful: If I have some money and after increasing it by 20% I have \$60, this does not mean that to find the original amount I subtract 20% of 60. What actually happens is that $60 = x(1 + 0.20)$ so by some algebra $x = 50$, but you would get $x = 48$ if you did it the naive way. If you add 20% to 48, you get 57.6 which is not 60.

We can have compound percentages: If I increase by a further 25% I would have \$75 and if I increase that by 20% I will have \$90. In general, if I increase by k% n times, this means I multiply by $1 + \frac{k}{100}$ n times so I multiply by $(1 + \frac{k}{100})^n$.

2 Geometry

2.1 Angles in polygons

We can have shapes and angles. Angles have that 360 degrees is a full rotation, 180 is a half rotation and 90 is a quarter rotation. We can have shapes like triangles (3 sides), quadrilaterals (4 sides, a square is a quadrilateral with all sides and angles equal), pentagons (5 sides), hexagons (6 sides), etc.

Proposition. The angles inside a shape (where by 'shape' I mean points connected by straight lines where no two of the lines intersect, ie a polygon) add up to $180(x - 2)$ where x is the number of sides in the shape.

Proof. We will only do the proof for the case that the polygon is convex (ie, has no dents or inward curves) because it is simple, but the general proof is deferred to level 4. I will prove the convex using just a picture.

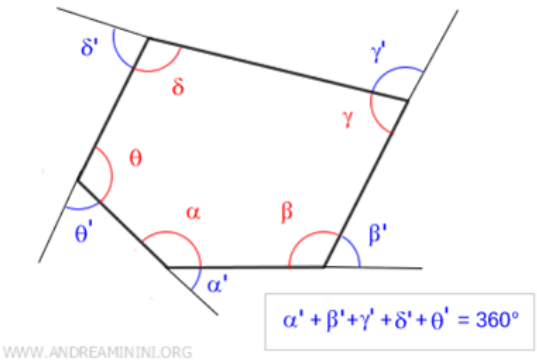


Figure 1

We see from figure 1 that the total of the red and blue angles is clearly 180 degrees times the number of vertices or sides, and that the sum of the blue angles is clearly 360 degrees as you could imagine dragging α' over to β' then those two together over to γ' and so on and then we will see that the total of the blues is clearly 360 degrees. The total of the reds is thus the total of everything (which is 180n) minus the total of the blues (which is 360), so it is $180n - 360 = 180(n - 2)$.

□

Definition. A regular polygon is a polygon where all its sides and angles are the same.

Example. Suppose the angles in a regular polygon are 165 degrees each and we want to find how many sides are in the polygon.

Then we know the sum of the angles is equal to 165n where n is the number of sides, since there are n angles each with size 165. We also know this total is $180n - 360$ by the formula above. Therefore these must be equal to eachother. We have applied the general principle of seeking two different pieces of information and now we will combine them. $165n = 180n - 360$ can be solved for n to get n=24.

Here are some definitions about angles to know.

Definition. An acute angle is an angle that is less than 90 degrees.

Definition. A right angle is an angle that is exactly 90 degrees.

Definition. An obtuse angle is an angle that is more than 90 but less than 180 degrees

Definition. A reflex angle is an angle that is more than 180 degrees and less than 360 degrees

2.2 The pythagorean theorem

If a triangle has an angle that is 90 degrees, ie a right angle, and the length of the two shorter sides are a and b, and the length of the longer side is c, then the length of the sides are related by $a^2 + b^2 = c^2$. This is the pythagorean theorem. I will show an example of how this works and then show a proof.

Example. Suppose we want to find the length of the bottom side in the following triangle (figure 2).

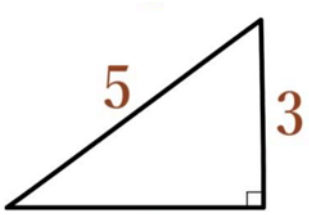


Figure 2

Lets call the length of the bottom side x. By the pythagorean theorem, $3^2 + x^2 = 5^2$. This implies that $x^2 = 16$. It turns out that $x = 4$ works. $x = -4$ also works but the length is never negative.

Definition. The area of something is how much space it fills. We define the area of a square with side length 1 to be 1.

Proposition. Suppose a triangle is rotated to have one side horizontal. Then we call this side the base. Suppose the length of the base is b and the height is h, then the area of the triangle is given by $\frac{1}{2}bh$.

Proof. The area of the orange and green parts combined in figure 3 is clearly equal to the base times the height. The area of the orange triangle is clearly half of that.


Figure 3

Figure 4 shows why this works when the triangle does not neatly fit into a rectangle as in figure 3.

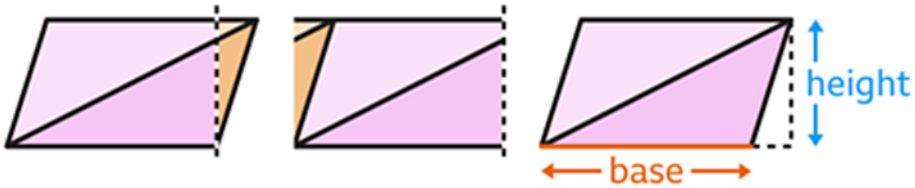


Figure 4

□

Now, as promised, I will prove the pythagorean theorem.

Theorem. If a triangle has an angle that is 90 degrees, ie a right angle, and the length of the two shorter sides are a and b, and the length of the longer side is c, then the length of the sides are related by $a^2 + b^2 = c^2$.

Proof. Again, we will do this with a picture.

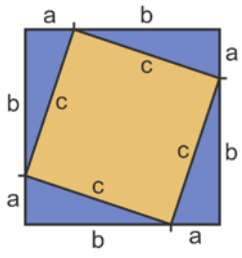


Figure 5

In figure 5, there are four blue triangles with sides abc. The area of the large square is the area of the orange square plus four of the blue triangles. This is $4(\frac{1}{2}ab) + c^2$. The area is also equal to the square of the side length, ie $(a + b)^2$ which can be expanded to obtain $a^2 + 2ab + b^2$. Therefore we know that two things equal to the area of the large square are equal to eachother, which means that $a^2 + b^2 + 2ab = c^2 + 2ab$ so therefore $a^2 + b^2 = c^2$.

□

This theorem can be generalized: If I move 3 meters forward and 4 meters to the right then the square of how much distance in meters I will have covered if I draw a line from my initial to final positions is $3^2 + 4^2$. Then if I move 12 meters up, since the up direction is perpendicular (ie makes a right angle) to any movement I do along the ground, it means that the square of the distance I have moved in total will be $3^2 + 4^2 + 12^2$, so it just so happens I will have moved a total of 13 meters if I had gone in a straight line. Therefore, in the three perpendicular directions (up, forward and right), I can add the squares of the distance each of my "coordinates", ie my height, rightness, and forwardness have changed to get the square of my overall distance. Alternatively I can find that long the diagonal of a $3 * 4 * 12$ brick is 13 in length by the same principle

Definition. The volume is the analogous thing to area for 3 dimensional things.

2.3 Pi

The definition of π or pi (which is about 3.14159265358979323846264338327950288419716939937510582097494459230781640628620899) is the distance around a circle that is 1 long.

In case this definition is not clear, Figure 6 shows what lengths I am referring to

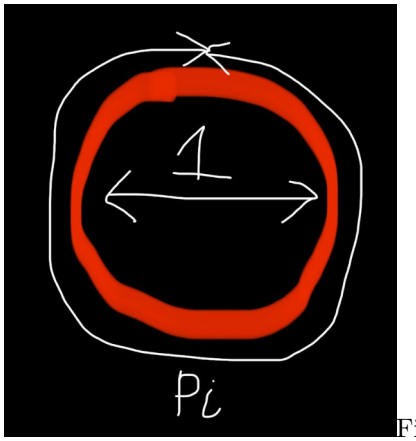


Figure 6

Definition. The circumference of a circle is the distance around it.

Definition. This is really two definitions. The radius of a circle (r) is half its length (diameter).

Now π can be defined as $\frac{C}{D}$ where C is the circumference and D is the diameter of any circle.

Then the area of a circle is given by $\pi * r^2$. The proof is reserved for level 3. Note that the area is clearly a multiple of r^2 as if we scale the circle the area changes related to the square of how much we scale the circle by, as it scales by that amount by the base and the height directions.

2.4 More on angles

Here are a few useful tools to find angles. Figure 7 speaks for itself.

Note that in figure 7 the horizontal lines are parallel.

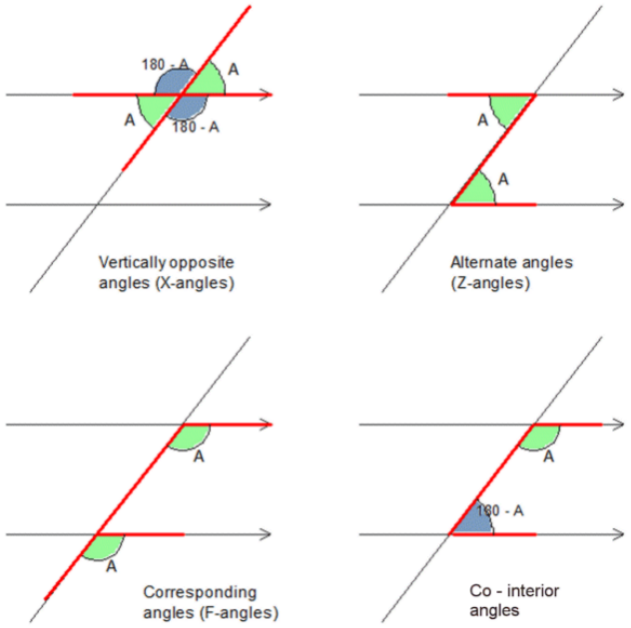


Figure 7

3 Prime factorization

Every positive whole number (except the special case of 1) can be uniquely factored into primes (apart from the order of the factors), for example $630 = 2 * 3 * 3 * 5 * 7$. This is because you can decompose the number until you are down to primes. The proof that this can be done uniquely is reserved for level 4. Recall that a prime number is a number with no factors other than 1 and itself. You can check that the first few primes are 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, ... Given a number, you can try dividing it by each prime until you can't anymore and have a factorization into primes.

4 Sets

You can also have sets of things. As an example, let A=$\{2,3,5,7\}$ be a set and B=$\{1,3,9\}$ be a set. Then we can do A$\cup$B to mean "A union B" which means the set of everything in A or B, which in this case would be $\{1,2,3,5,7,9\}$. A$\cap$B means "A intersection B" meaning everything in both A AND B, which in this case would be $\{3\}$. The notation $3\in$A means 3 is an element of A. The notation $\{3,5\}\subseteq$A means $\{3,5\}$ is a subset of A, ie it is contained in A. We can do venn diagrams.

$\mathbb{N}$ is the symbol for the set of positive integers or natural numbers. Convention about if this includes 0 varies. $\mathbb{Z}$ is the symbol for the set of all integers (not necessarily positive). $\mathbb{Q}$ is the symbol for the set of rational numbers (numbers that can be represented by fractions of integers, these are not all numbers as we prove numbers like π and some square roots are not rational). $\mathbb{R}$ is the symbol for the set of all real numbers. A\B is typically used to mean the set of stuff in a set A but not a set B.

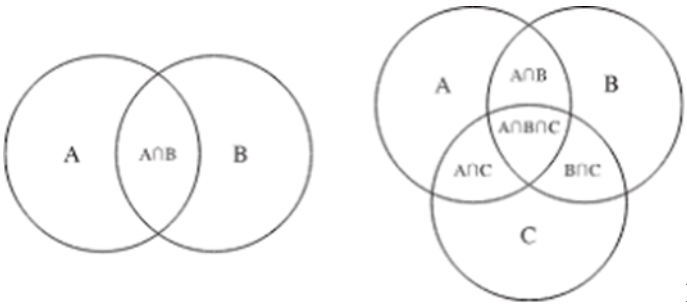


Figure 8

You can also write things in venn diagrams to illustrate exactly which sets some elements are in or not in. Figure 8 shows a venn diagram.

To compute the lowest common multiple or highest common factor of two numbers (Lowest common multiple meaning the smallest number that is a multiple of both, for example the LCM of 30 and 40 is 120, and highest common factor means the largest number that is a factor of both, so for example the HCF of 30 and 40 is 10) you can put the prime factors into a venn diagram and multiply together everything in the diagram and everything in the intersection respectively. This works essentially because to be a common multiple you need to have all the prime factors of both numbers, similar for common factors.

5 More algebra

5.1 Simultaneous equations

It is possible to solve simultaneous equations.

Example. Suppose we have $3x + 4y = 46$ and $2x + y = 19$, then we can rearrange any equation to get one variable, in this case that $y = 19 - 2x$. Then we can put that back into the other equation to get $3x + 4(19 - 2x) = 46$, and we can solve this for x to get $x = 6$, then by either equation, say the second one $2 * 6 + y = 19$, so $y = 7$. In general, systems of multiple equations, or even single equations, just anything in general, can possibly have one single solution, more than one, infinitely many, or zero solutions, so be careful.

5.2 Inequalities

We can also solve inequalities.

Example. if we want to find the set of values of x such that $2x + 3 < 5$, we can rearrange to get $2x < 2$ and then x<1. It is ok to multiply or divide inequalities by positive real numbers or add or subtract real numbers from both sides, but if you multiply both sides by a negative real number you must flip it. Ie, if we have $3 - x < 2$ this implies $-x < -1$, so because minus x lies to the left of -1 on the number line, we have $x > 1$.

Remarks on inequalities:

1. Notice that if x is positive, $\frac{1}{x}$ is smaller the larger x is – The more slices you cut a cake into the smaller each slice is. This means that if $a < b$, $\frac{1}{a} > \frac{1}{b}$, so if you take the reciprocal of both sides of an inequality where you know both sides are positive (or equivalently, both negative) you need to flip it. Note that one divided by a number is called the reciprocal of that number.

2. If x is positive x^2, or any positive power of x in fact, gets bigger as x does, which I'm sure does not need much convincing. Therefore if both sides of an inequality are positive we can safely raise them both to a positive integer power (in fact any positive power, but we have not defined this yet, and it is less obvious that it is increasing but after level 3 we could argue that with differentiation)

Inequalities can be strict or not strict. $<$ means less than and $\leq$ means less than or equal to and similarly for greater than. So for example, $5 \leq 5$, but it is not true that $5 < 5$.

5.3 Graphs

We can plot equations on graphs. Play around with desmos to see how this might work. We can have a graph like $y = 3x + 5$, then for every change in x by 1 y will change by 3, and when x is 0 y will be 5, so we have a straight line with "slope" 3 that intersects the y axis (the line $x = 0$) when $y = 5$, and by some algebra we find that $y = 0$ when $x = -\frac{5}{3}$. The image below shows the graph.

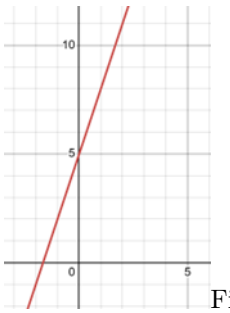


Figure 9

Figure 9 shows the graph of $y = 3x + 5$.

The simultaneous equations above can be interpreted as graphing 2 lines and finding where they meet.

We can find a line between 2 points.

Example. Lets say we want to find an equation for the line that passes through $(2, 3)$ and $(5, 7)$ where this means the point $x = 2$, $y = 3$ and the point $x = 5$, $y = 7$. We know that any line equation has the form $y = mx + c$ (however a vertical line cannot be made into an equation of this form and instead will be an equation of the form $x = c$). Therefore $3 = 2m + c$,$7 = 5m$+c. Solving for m and c gives that $m = \frac{4}{3}$,$c = \frac{1}{3}$. So the equation is $y = \frac{4}{3}x + \frac{1}{3}$. Rearranging this gives $3y - 4x - 1 = 0$. Note that the convention is to multiply by common denominators so that everything is an integer, and then divide through by any common factors if necessary.

5.4 Quadratic equations

Definition. A quadratic is an expression that looks like $Ax^2 + Bx + C$, ie it involves squaring a variable or multiplying it by a number or adding it to a number but that is it.

They can be factorized and solved. As an example, suppose we want to find all the values of x such that $x^2 - 5x + 6 = 0$. By some algebra, $x^2 - 5x + 6 = 0$ implies that $(x - 2)(x - 3) = 0$ (expand this, I will talk later about how to actually find this factorization), so x is 2 or 3. Therefore the solutions are $x = 2$ and $x = 3$, as $(x - 2)(x - 3)$ is 0 exactly when either $x - 2$ or $x - 3$ is 0. The solutions to $x^2 - 5x + 6 = 0$ are called the roots of $x^2 - 5x + 6$ but such equations do not always factor easily. An example of an equation that is not so easy to factor like above is $x^2 + 2x + 2 = 0$.

Note that, as with the example $x^2 - 5x + 6 = (x - 2)(x - 3)$, it is generally true that$(x - A)(x - B) = x^2 - (A + B)x + AB$, as we can just expand it to verify this fact. This tells us that we should look for numbers that add up to minus the coefficient (ie the thing multiplying) of x and multiply up to the term not depending on x (ie the constant term). These may not be positive, for example in $x^2 - x - 2 = 0$ the numbers I am looking for are 2 and -1. Usually you want to find these by trial and error, otherwise you want to use a formula which I will show later.

6 Probability and statistics

We all have an intuitive idea of what probability is. You can think of "The probability of X happening is 0.6" to mean that if you keep doing trials, you would expect X to happen on about 60% of trials, since 0.6 is analogous to 60%. We can multiply probabilities, X happens twice in a row with probability $0.6 * 0.6$, 3 times in a row with probability $0.6 * 0.6 * 0.6$, in general if an event has probability p, it happens n times in a row with probability p^n (as long as each of the n times are independent of eachother). p is always between 0 and 1 for this to make sense. Also, the probability of something not happening is 1-p: An analogy is if there is a 70% chance of rain, this translates to $p = 0.7$, then the chances of not rain is 30%, which translates to $p = 0.3$, which indeed is the same as $1 - 0.7$.

You can also add probabilities provided events are mutually exclusive. What mutually exclusive means is that they cannot

both happen. For example, a dice cannot land on both a 1 and a 2. The probability it lands on a 1 is $\frac{1}{6}$ and the probability it lands on a 2 is also $\frac{1}{6}$. So the probability it lands on a 1 or a 2 is $\frac{1}{6}+\frac{1}{6}=\frac{1}{3}$. However, you cannot say that the chance of getting a 1 on either the first roll or the second roll is $\frac{1}{3}$, since this is false. It is actually the sum of the probability that you get it on the first roll with the probability that you do not get it on the first roll but get it on the second roll. These are actually mutually exclusive. You will end up getting $\frac{1}{6}+\frac{5}{6}*\frac{1}{6}=\frac{11}{36}$. We can multiply probabilities if we want to get the probability of both of the two things happening because the events do not depend on one-another, i.e. they are independent.

Probability is an area with a lot of mathematical misconceptions that I have seen from teachers and wikipedia articles. Ask yourself what is the probability that you get the same number on a dice twice in a row. Naively you might say that it is $\frac{1}{6}*\frac{1}{6}=\frac{1}{36}$, but this is false, as what happens is that what you get on the first roll does not really matter and then what you get on the second roll merely has to be the same, and regardless of what you got on the first roll this has probability $\frac{1}{6}$.

In statistics, if I have a list of data, there are a few ways to try to get an average or an idea of the spread. Lets say my data is 1,3,4,5,6,6,6, then the number that occurs in the middle of the sorted data (median) is 5, the number that occurs most often (mode) is 6, the result of adding them and dividing by the number of data (mean) is about 4.43, and the range (largest minus smallest) is 5. We can also calculate ranges excluding the most extreme 25% or 5% of data on either side, there are different conventions but the idea is to have extreme values (outliers) be excluded.

Example. If our data is simply $0,1,2,3,4,...,98,99,100$ then the "most extreme" 5% on each side will be the numbers from 0 to 4 and from 96 to 100 so they will be excluded. I'm not sure about the standard convention for the exact amount removed but just the percentage of the number of data points and round it to the nearest integer and that should be good enough, escpecially for large data sets.

7 Trigonometry and more geometry

7.1 Trigonometry

In a right angled triangle with another angle x, we have the long side (h), the other side that touches the angle (a) and the remaining side o. Then the ratio $\frac{o}{h}$ is the **sine** of the angle x, which we write $\sin(x)$. Similarly we write $\frac{a}{h}$ as $\cos(x)$ (**cosine**) and $\frac{o}{a}$ as $\tan(x)$ (**tangent**). These are trigonometric functions.

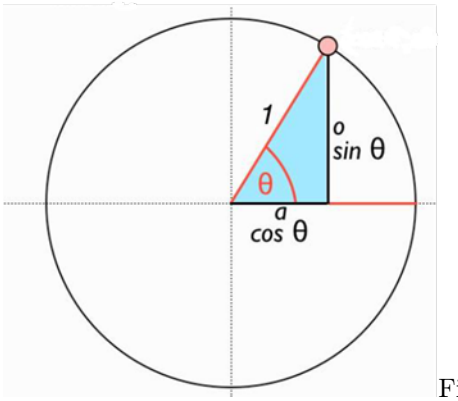


Figure 10

Figure 10 shows an interpretation of this definition as coordinates along a circle with radius 1. We see that if $h=1$ then $\cos(\theta)$ and $\sin(\theta)$ are exactly our x and y coordinates when we move by an angle θ along the circle anticlockwise from the point (1, 0).

Now we can write $\sin^{-1}(x)$, or the notation I will use is $\arcsin(x)$ to mean the angle from -90 to 90 degrees such that its sine is equal to x. We can do the same for $\arccos(x)$ where we pick the angle from 0 to 180 degrees such that its cosine is equal to x, and we can do the same for $\arctan(x)$ to pick the angle from -90 to 90 degrees such that its tangent is equal to x (although its range does not include the endpoints). The circle interpretation is the way that sin, cos and tan can be defined for negative angles. When we see the graphs of these functions, it will make sense why I have chosen these specific ranges of degrees.

By the circle interpretation, $\sin(0°)=0$ and $\sin(90°)=1$ and $\cos(0°)=1$ and $\cos(90°)=0$, and we have a useful formula:

$\tan(x) = \frac{o}{a} = \frac{o/h}{a/h} = \frac{\sin(x)}{\cos(x)}$ which we can use to show that $\tan(0°) = 0$. Note that $\tan(x)$ is undefined when $\cos(x) = 0$.

We can get more values of sin, cos and tan at specific values, but first we need to talk about how to deal with square roots, so we will come back to this.

7.2 More geometry stuff

Definition. Triangles are similar if they have the same angles, meaning you can transform one into the other by scaling, rotating, reflecting and moving it. They are congruent if you impose also that their sides are the same.

Equivalently, similar triangles have corresponding sides in the same ratio. Congruent triangles have the same shape and size.

Definition. Two lines are said to be parallel if they point in the same direction

Definition. A parallelogram is a 4 sided shape with 2 sets of parallel sides, as shown in figure 11.

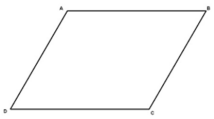


Figure 11

Proposition. The area of a parallelogram is just the base times the height

Proof. You can just turn it into a rectangle with the same base and height by cutting it as in figure 12.

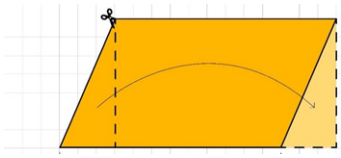


Figure 12

□

Definition. A trapezium is a 4 sided shape with one set of parallel sides (whether this means exactly one or at least one depends on the convention). Figure 13 shows an example of one.

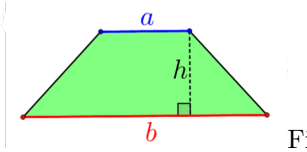


Figure 13

Proposition. Suppose that a trapezium has sides labelled as in figure 13. Then its area is given by $h(\frac{a+b}{2})$

Proof. Just split it into 2 triangles with the cut shown in figure 14 and find the areas using a previous proposition on the area of a triangle and add them together.

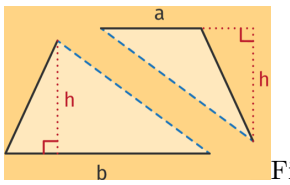


Figure 14

□

An equilateral triangle (all sides and angles the same) will have all its angles 60 degrees since all three of its angles must add up to 180 degrees by a formula above. If we take half of it by drawing a line from the top vertex to the exact middle of the base and cut along that line, we will therefore get a triangle with angles 30, 60 and 90 degrees. Say we start with an equilateral triangle with side lengths 1 with one side horizontal then cut it so we only have the left side. Then the base will have length 1/2, then the side that was originally part of the equilateral triangle that is on the left will be unchanged and have length 1. Lets call the remaining side length x. By pythagoras, $x^2 + (\frac{1}{2})^2 = 1^2$, so by a bit of simplifying we have that $x^2 = \frac{3}{4}$. This motivates what we will do next.

8 Square roots

Definition. The square root of a positive number ($\sqrt{x}$) is the positive number that gives x when you square it. Note that we only define it for $x \geq 0$ as no real number squares to give a negative number.

We see that in the example given at the end of the last section, $x = \sqrt{\frac{3}{4}}$. Note that the square root of a number x can be written as $x^{\frac{1}{2}}$ because by the usual rules of powers $x^{\frac{1}{2}}x^{\frac{1}{2}} = x^{\frac{1}{2}+\frac{1}{2}} = x^1 = x$. The fact that we can do this is something we will make more rigorous in level 4 where we will define powers formally for real numbers and show that they satisfy these rules for real numbers and not only for integers (ie, numbers like ...,-2,-1,0,1,2,... and not like 0.73) and we will know that we have a proper definition, but for now you should assume that the rules of exponents are consistent and that we can "abuse" them in this way for real numbers. Now note also that $x = \sqrt{\frac{3}{4}} = (\frac{3}{4})^{\frac{1}{2}} = \frac{3^{\frac{1}{2}}}{4^{\frac{1}{2}}} = \frac{\sqrt{3}}{2}$ by exponent rules and the fact that the square root of 4 is 2. Now we have all the side lengths of our triangle and can therefore deduce the following from what we did in the previous section:

$$\sin(30°) = \cos(60°) = \frac{1}{2}$$

$$\cos(30°) = \sin(60°) = \frac{\sqrt{3}}{2}$$

$$\tan(30°) = \frac{1}{\sqrt{3}}$$

$$\tan(60°) = \sqrt{3}$$

Now in a 45, 45, 90 triangle the two sides touching the right angle are the same length, and by pythagoras if these sides have length 1 then the remaining side has length $\sqrt{2}$. Therefore we have

$$\cos(45°) = \sin(45°) = \frac{1}{\sqrt{2}}$$

$$\tan(45°) = 1$$

Note that negative numbers don't have square roots because you cannot square a real number and get a negative number, since negative times negative is positive, zero times zero is zero, and positive times positive is positive. However, notice that I've been saying real number and not number the entire time. This is because we can allow for square roots of negative numbers and get "imaginary numbers", but we will not discuss this point further in this level.

9 Functions

9.1 Definition

We can make functions of x. A function is basically a rule to take a number and output another number, but number can be replaced with anything in the definition as we'll see in later levels. For example, if we define a function as $f(x) = x^2$ then $f(3) = 9$. A function is something, usually denoted f, that takes in one value and returns exactly one value.

The set of values such that a function is defined is called its domain and the set of values it can return is called its codomain.

The range of a function is not the set of values that a function can return but the set of values that are actually returned. In the case above we can define $f(x) = x^2$ with domain real numbers, codomain real numbers and range non-negative numbers (since the square of a number is never negative as whether you square zero or a negative thing or a positive thing you get a non-negative thing). It is best practice to specify the domain of a function when you define it.

9.2 Graphs

We can compute a few values of x^2 and put these on a graph, figure 15 shows what the result looks like.

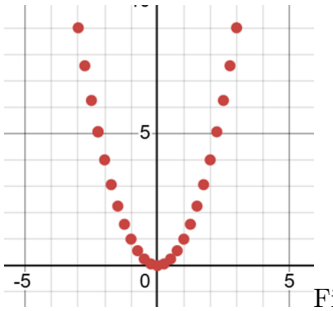


Figure 15

Given these points gives a rough idea of what the graph will look like when we sketch it properly, and it gives enough information that you could draw a sketch. (The actual graph shown in Figure 16)

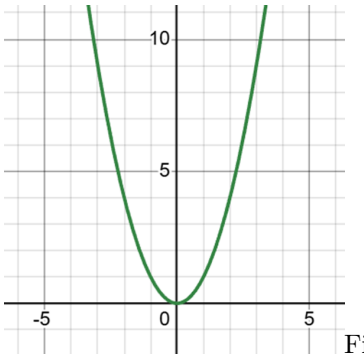


Figure 16

This is the graph of the function $y = x^2$.

This shape is called a parabola. In general you can compute some values and try to connect the points in order to graph any sensible function. For example, figure 17 shows the graph of $y = x^3$.

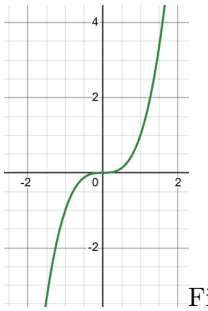


Figure 17

Once you have seen these graphs once you will know the general shape and will eventually be able to sketch them without having to compute values and connect the dots. We will see how this works as we go on.

We can transform objects, in particular graphs. Here are some ways we can do that:

- Shifting (Translation): This preserves lengths and angles

- Reflection about a line: This preserves lengths and angles but not orientation

- Enlargement by an axis: This does not preserve lengths or angles

- Enlargement by both axes by the same factor on both axes: This preserves angles but not lengths

If $y = f(x)$ and we know what that graph looks like, then we can work out what the graph $y = f(x + a)$ must look like

for any choice of a. This is because the value of the function $f(x+a)$ at any x value is just the same as the value of $f(x)$ at the x value a distance of a units away.

Figure 18 shows some examples.

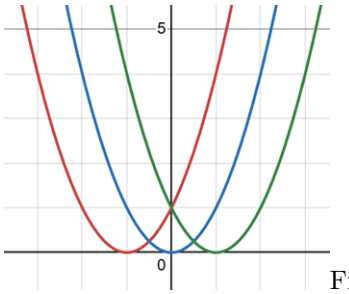


Figure 18

In figure 18:

Red graph is $y = (x+1)^2$

Blue graph is $y = x^2$

Green graph is $y = (x-1)^2$

Note that if we take $y = f(x) + a$ we are shifting y by a, so that means the graph will shift upwards if a is positive and downwards if a is negative. As usual, we will show this on a diagram.

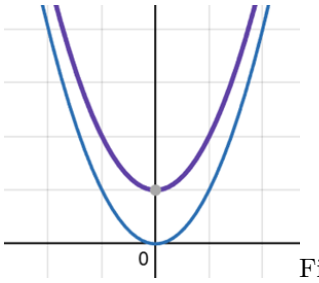


Figure 19

In figure 19, Blue is $y = x^2$, Purple is $y = x^2 + 1$.

We can also have $y = af(x)$,$y = f(ax)$, I will show both of those in Figure 20, then after that one can combine transformations to get graphs of $y = af(bx + c) + d$ for any choices of a, b, c and d.

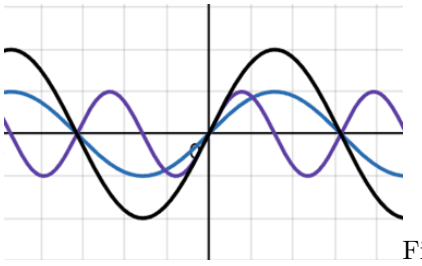

Figure 20

Figure 20 may look intimidating, but just look at one graph at a time. We will say that the blue graph is $y = f(x)$, then purple is $y = f(2x)$ (For each x value we need to find f of double that so it moves along twice as fast), and black is $y = 2f(x)$ (Each y value is doubled from blue).

Those are all the simple transformations on graphs.

When we learn how to complete the square it will follow how you can use this to sketch any graph of the form $y = ax^2+bx+c$.

Lets now sketch the graphs of $sin(x)$,$cos(x)$ and $tan(x)$.

As you move around a circle, it is easy to see that your x and y coordinates which are cosine and sine of the angle respectively if you plot them on the y axis with time on the x axis will look like this:

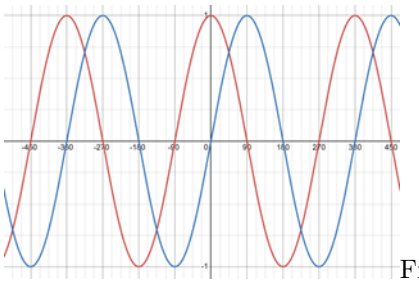


Figure 21

In figure 21, we see the number of degrees labelled on the x-axis. In this figure, the graph of $y = \sin(x)$ is the blue one and the graph of $y = \cos(x)$ is the red one.

As usual it is possible to transform these graphs as above.

To find the graph of $\tan(x)$, note that $\tan(x) = \frac{\sin(x)}{\cos(x)}$ we want $\frac{y}{x}$ as we move along the circle, ie the slope of the line from the origin to that point on the circle. Figure 22 shows a graph of how that changes as we move around the circle:

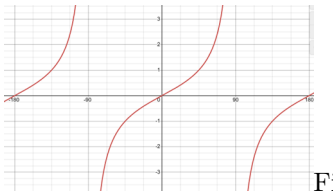


Figure 22

Again it is possible to do transformations on this.

Now notice how in each of these graphs of functions, all possible y values (outputs of the function) can be achieved by infinitely many x values. This moves me to a very important general point:

10 Things to be careful of

Do not cancel zeroes, for example $0 * 3 = 0 * 5$ does not imply $3 = 5$. If cancelling any factor, justify that it is not zero. If you have $ac = bc$ and cancel the c factor, what you should do is deal with the $c = 0$ case separately, ie $ac = bc$ implies that either $a = b$ OR $c = 0$.

Do not cancel squares and square roots when the square is inside the square root unless you justify that the thing being squared inside the square root is a non-negative real number. For example, $\sqrt{(-3)^2} = \sqrt{9} = 3$ which is not equal to -3. This is because square root is defined the positive square root. If it were a proper inverse of the square function, it would have to take two values (for example $\sqrt{9} = (3, -3)$) and then it would not be a function, by definition, since a function takes exactly one value.

If you square both sides of anything, you must either be sure that it is not the case that one side is non-zero but minus the other side (Since then they could be non-equal but have equal squares, both sides being non-negative reals would suffice), or check after for extraneous solutions

Example. Suppose that $x - 1 = 3$. Clearly $x = 4$, but for the sake of example I will square both sides to show what goes wrong.

$$(x-1)^2 = 9$$
$$x^2 - 2x + 1 = 9$$
$$x^2 - 2x - 8 = 0$$
$$(x-4)(x+2) = 0$$

So the extraneous solution $x = -2$ arises. It is true when squaring both sides that each step is implied by the previous, but that does not mean that each step is equivalent to the previous.

The reason is essentially that more things can be squared to give 9 than just 3. If a function is one-to-one from the domain to the range, you may apply it to both sides or cancel it from both sides, but otherwise it requires justification.

Just like how squaring both sides can cause you to gain solutions, the opposite is true as you can lose solutions by unsquaring both sides. For example, $x = -3$ is a solution of $x^2 = 3^2$ but naively unsquaring both sides would just give $x = 3$. This is because, again, "unsquaring" is inherently a multivalued process.

When cancelling squares on both sides or cancelling squares on one side or square rooting the other, you should put a $\pm$ on one of the sides to ensure no loss of information. This symbol means "plus or minus". We would say that $x^2 = 3^2$ implies $x = \pm 3$.

Also, note that trigonometric functions are many-to-one over the real numbers, so similar problems arise. If $y = \arcsin(x)$, it is true that $x = \sin(y)$ meaning $x = \sin(\arcsin(x))$ when x is from -1 to 1 inclusive. However, if $y = \sin(x)$ it is not necessarily the case that $x = \arcsin(y)$, because consider what happens if $x = 180°$: arcsin is defined to take the value from -90° to 90°, so $\arcsin(\sin(180°)) = 0$ which is a counterexample, kind of like how the square root of a negative number squared is a positive number since square root is defined to output positive values.

From $A = B$, one may always infer $f(A) = f(B)$. To infer $A = B$ from $f(A) = f(B)$, f must be one-to-one on the relevant domain.

I guess my point is you need to carefully justify cancelling functions that you think are inverses of each other or applying or cancelling one-to-many functions to both sides of an equation.

Lets do an example where we are being careful to really drive this home.

Example. We want to solve for x in $\sqrt{3x-5} - \sqrt{x+6} = 1$ for real values of x such that the above expression is real (we need $x \geq \frac{5}{3}$ for this because of the first square root and the fact that we don't want to take the square root of a negative number, so that is our domain).

Note that square roots of real numbers are always positive, so here is one way to do that.

Lets square both sides, remembering to check each solution at the end to make sure it is not extraneous. This gives $3x - 5 + x + 6 - 2\sqrt{(x+6)(3x-5)} = 1$, since we know it is true that $\sqrt{x+6}\sqrt{3x-5} = \sqrt{(x+6)(3x-5)}$ by exponent rules. Rearranging this a little bit gives

$$4x = 2\sqrt{(x+6)(3x-5)}$$
$$2x = \sqrt{(x+6)(3x-5)}$$

Now lets square both sides again

$$4x^2 = (x+6)(3x-5) = 3x^2 + 13x - 30$$
$$x^2 - 13x + 30 = 0$$
$$x = 3, x = 10$$

But one can check by plugging in $x = 3$ to both sides of the equation that it is extraneous and only the solution $x = 10$ works. The problem was when we squared both sides the first time. Squaring both sides the second time was fine because square roots of real numbers are always positive by convention and the domain is $x \geq \frac{5}{3} > 0$ so when we squared a square root we were squaring something we knew couldn't be negative, so we were applying a function that was actually one-to-one on the domain of its possible values.

11 Sequences

A sequence is a list of numbers. If I give you an expression for the n'th term such as $n^2 + 1$ you can generate your sequence by putting 1, 2, 3, etc into the expression. For the above expression the resulting sequence would be 2, 5, 10, 17, 26, 37, etc

Now if I give you a sequence where the rule to get from one term to the next is something simple, such as adding 4, you can reverse engineer the rule by pattern spotting. For example, a sequence may be given by 1, 5, 9, 13, 17, 21, ... A common mistake I've seen is to say that "$n+4$" is the expression that returns the n'th term in this sequence. But that is the rule to get from one element to the next which is not the same thing. The sequence generated by $n+4$ would actually be the sequence 5, 6, 7, 8, 9, ... For a sequence where each term is, say 4, more than the previous, trying 4n gets us close – this gives us the sequence 4, 8, 12, 16, 20,... and each term is 3 too large, so 4n-3 gets us the desired sequence 1, 5, 9, 13, 17, 21, ...

Trivia: The above paragraph was the first question on my Maths GCSE.

12 Infinity

It is here that I will briefly explain the concept of infinity since it is a confusing concept to grasp and I don't expect that this brief explanation will have you fully understanding it but hopefully as we go on in our maths journey you will get a grasp of it.

Basically, infinity is not a number in the traditional sense, however it makes sense to say that if I have a sequence like 1,10,100,10000,100000000,10000000000000000, etc, then the sequence is *tending towards* infinity. Infinity is a concept that you have likely heard of and sort of understand, and the reason we say it is not a number is because you cannot add it like you cannot say $\infty + 1$ or something, as infinity does not follow the rules of arithmetic. We can say "going off to infinity", and more importantly, it makes sense to say "there are infinitely many numbers" or "numbers go on forever".

13 More on functions

13.1 Inverse functions

We can go beyond square roots. We can have cube roots of numbers - $\sqrt[3]{x}$ is the cube root of x, ie the real number y such that $y^3 = x$. These are unique because for positive x, x^3 is increasing as x increases, and for negative x, $(-x)^3 = (-x)(-x)(-x) = -(x^3)$ since there are an odd number of minuses and pairs of minuses cancel, so x^3 is increasing for negative x as well, and from this and the graph $y = x^3$ you see that it is a one-to-one function that can map any real number to any other real number so it does have a proper inverse, ie the cube root. Similar to with square roots, $\sqrt[3]{x} = x^{\frac{1}{3}}$.

Inverse functions are a more general concept – a function you get by "undoing" another function. We have seen examples like arcsin as an inverse of sin, but let's see how you can find inverse functions in general. If $y = f(x)$ we want to find x as a function of y, and this function we call $x = f^{-1}(y)$. As an example, if $f(x) = 3x + 7$ here is how we can find an inverse (whenever the function is one to one from its domain to its range):

$$y = 3x + 7$$

$$3x = y - 7$$

$$x = \frac{y-7}{3}$$

So our rule for f^{-1} is that $f^{-1}(x) = \frac{x-7}{3}$, where I have written x in place of y in $f^{-1}(y) = \frac{(y-7)}{3}$. Let's see geometrically what this does:

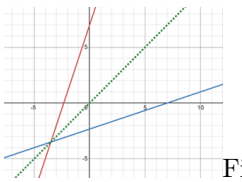


Figure 23

Figure 23 shows $f(x)$ in red and $f^{-1}(x)$ in blue. We note that if y is an inverse function of x it is essentially like x as a function of y, so it is essentially like swapping the axes around, which has the geometric effect of reflecting about the green dotted line $y = x$.

13.2 Composite functions

We also can have compound functions where we essentially do 1 then the other.

Example. Lets define two functions $f(x) = 3x + 7$ and $g(x) = x^2 + 1$, then:

$f(g(x)) = f(x^2 + 1) = 3(x^2 + 1) + 7 = 3x^2 + 10$

$g(f(x)) = g(3x + 7) = (3x + 7)^2 + 1 = 9x^2 + 42x + 50$

Notice that they are not the same.

14 Powers and roots

When dealing with problems involving powers, it is usually a good idea to put everything on the same base, meaning the number on the "base" of the power should be consistent throughout the equation. I will demonstrate this by an example.

Example. Lets try to simplify $\frac{3^5 * 27^{\frac{1}{4}}}{\sqrt[3]{9}}$.

First, we spot that $\sqrt[3]{9}$ is $9^{\frac{1}{3}}$, then we notice that our bases are 3, 9 and 27, which are 3^1,3^2,3^3 respectively, so we write $\frac{3^5 * (3^3)^{\frac{1}{4}}}{(3^2)^{\frac{1}{3}}} = \frac{3^5 * 3^{\frac{3}{4}}}{3^{\frac{2}{3}}}$. by a rule of powers, and if we multiply things with the same base we can add powers, we know how to add fractions from level 1, and we end up with $3^{\frac{61}{12}}$ as our answer.

Example. Suppose we want to solve for x in $2^{3x-5}4^{1-x} = 1$.

Then again we want to get everything onto the same base, and in this case 2 is the obvious choice. We can write $2^{3x-5} * (2^2)^{1-x} = 1$ and use rules of powers to get $2^{3x-5}2^{2(1-x)} = 1$ so $2^{3x-5+2(1-x)} = 1$. Since something that is not 1 to the power of a real number being 1 implies the thing in the power is 0 (since if we increase the power the result strictly increases so the result can only be 1 once), we therefore have $3x - 5 + 2(1 - x) = 0$ which after simplifying implies $x = 3$.

Example. We want to simplify $\sqrt{80} + \sqrt{125}$.

We do this by fully factoring the square roots into primes. This gives $\sqrt{5 * 2^4} + \sqrt{5^3}$. We now pull out all the squares we can, ie for powers that are larger than 2, we split them until each power is at most 2: $\sqrt{(5 * 2^2 * 2^2)} + \sqrt{(5 * 5^2)}$. Now we take the squares out of the square roots using rules of powers: $\sqrt{(2^2)}\sqrt{(2^2)}\sqrt{5} + \sqrt{(5^2)}\sqrt{5}$. Since we are assuming that we want the positive square root, we can cancel the square root with the square. Then $2 * 2 * \sqrt{5} + 5 * \sqrt{5} = 9\sqrt{5}$ is our final answer.

15 Lower and upper bounds from rounding

Example: $\frac{3.2+7.5*9.1}{6.4-3.7}$ is a fraction where everything is rounded to one decimal place and we want to find the largest and smallest possible value it can take. Note that since something is larger the smaller the denominator is whenever the denominator is positive, to maximize it we want to maximize the numerator and minimize the denominator. As an example, 3.2 is rounded to 1 decimal place so its possible range of values is 3.15 to 3.25. So the maximum value of the fraction is

$\frac{3.25+7.55*9.15}{6.35-3.75} \approx 27.82$

The squiggly equals sign means approximately equal to. As to minimize the denominator we minimize the parts we are adding and maximize the parts we are subtracting. And the minimum possible value is

$\frac{3.15+7.45*9.05}{6.45-3.65} \approx 25.20$

Note that technically these maxima or minima may be best bounds that are not necessarily attained.

16 The quadratic formula

Now let's go back to quadratic equations, ie equations like $x^2 + ax + b = 0$ for varying values of a and b. Note that $(x + c)^2 = x^2 + 2cx + c^2$ by simple algebra, so any expression of the form $x^2 + 2cx + d$ just requires you to add or subtract

something and then it will be exactly $(x+c)^2$. So what we can do is $x^2+2cx+d = x^2+2cx+c^2-c^2+d$ since adding $+c^2-c^2$ does nothing, and therefore we have that $x^2+2cx+d = x^2+2cx+c^2-c^2+d = (x+c)^2+d-c^2$. In practice:

x^2+4x+5 has $2c=4$ so $c=2$ and $d=5$ so $x^2+4x+5 = (x+2)^2+5-2^2 = (x+2)^2+1$. This is called completing the square

Now let's use this to solve an equation that we may not be able to factor easily: $x^2-6x+12 = x+1$. First, move everything to one side, so we have $x^2-7x+11=0$. Completing the square (we solve for c and d,

$$c=-7/2, c^2 = \frac{49}{4}, d=11, c^2-d = \frac{5}{4}$$

and so $x^2-7x+11 = (x-\frac{7}{2})^2-\frac{5}{4}$, so we have that $(x-\frac{7}{2})^2-\frac{5}{4}=0$ so $(x-\frac{7}{2})^2=\frac{5}{4}$. Therefore $x-\frac{7}{2} = \pm\sqrt{\frac{5}{4}} = \pm\frac{\sqrt{5}}{\sqrt{4}} = \pm\frac{\sqrt{5}}{2}$, so $x = \frac{7}{2}\pm\frac{\sqrt{5}}{2}$

However, this does not always work: For $x^2+4x+5=0, (x+2)^2+1=0$ by the example above so $(x+2)^2=-1$ but as discussed no real number squared is -1.

Theorem. If $ax^2+bx+c=0$ then the possible values for x (ie the solution(s) to the equation) are given by the formula $x = \frac{-b\pm\sqrt{b^2-4ac}}{2a}$

Proof. Now we will derive the quadratic formula by taking the example above. Set $ax^2+bx+c=0$ and seek a formula for x. Note that on most school tests, you just need to be able to use the formula, not re-derive it like we will do here. Let's do some algebra.

$$ax^2+bx+c=0$$
$$x^2+\frac{b}{a}x+\frac{c}{a}=0$$

Assuming a is not 0, or else this is simple to solve. Now let's complete the square

$$(x+\frac{b}{2a})^2-(\frac{b}{2a})^2+c/a=0$$
$$(x+\frac{b}{2a})^2=(\frac{b}{2a})^2-c/a$$
$$x+b/2a = \pm\sqrt{\left(\frac{b}{2a}\right)^2-\frac{c}{a}} = \pm\sqrt{\frac{b^2}{(2a)^2}-\frac{4ac}{4a^2}}$$
$$= \pm\sqrt{\frac{b^2-4ac}{4a^2}} = \pm\frac{\sqrt{b^2-4ac}}{\sqrt{4a^2}} = \pm\frac{\sqrt{b^2-4ac}}{2a}$$

Note that actually $\sqrt{4a^2}$ is not $2a$ unless $a\geq 0$. However, it is $\pm 2a$ and since we already had a $\pm$ in the equation when we did this step, we absorb the new $\pm$ that comes from taking the square root and absorb it into the $\pm$ already present in the equation.

Therefore, rearranging, $x = \frac{-b}{2a}\pm\frac{\sqrt{b^2-4ac}}{2a} = \frac{-b\pm\sqrt{b^2-4ac}}{2a}$.

The quadratic formula is therefore

$$x = \frac{-b\pm\sqrt{b^2-4ac}}{2a}$$

as desired.

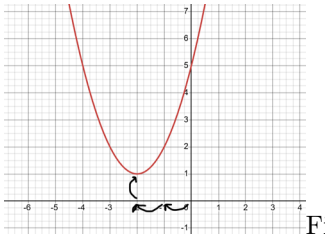
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Note that when solving an equation using this formula, a lot depends on the form of the number inside the square root b^2-4ac. If this is a square number, we will have nice solutions, otherwise we will not, and if it is negative we will not have real number solutions.

There is also a convention sometimes used with the $\pm$ symbol where if it appears twice in an equation it may be the case that they should either be both + or both -, but this is not always true. However, if the symbols $\pm$ and $\mp$ are both used in an equation or expression then they have strictly opposite signs.

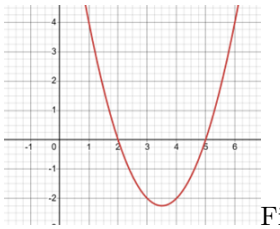
17 Quadratic graphs

Aside from deriving the quadratic formula, completing the square has another important use which is sketching graphs of quadratics. I will give some examples. We start with the typical parabola $y = x^2$. Now suppose we want to sketch the graph $y = x^2 + 4x + 5 = (x+2)^2 + 1$. We can sketch the graph $y = (x+2)^2$– we just have to take $y = x^2$ and shift it 2 to the left as instead of asking "what do we get when we square x" we are asking "what do we get when we square the thing 2 to the right of x on the number line" – recall graph transformations. Then we add 1 so we shift the graph 1 up. Figure 24 shows what this looks like.


Figure 24

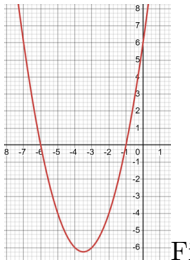
By this method we can sketch any graph of the form $y = x^2 + Ax + B$. We can also sketch something like $y = 4x^2 + 17x - 3$: Observe that $y = 4\left(x^2 + \frac{17}{4}x - \frac{3}{4}\right)$ and $x^2 + \frac{17}{4}x - \frac{3}{4}$ is a function which we can sketch with the above method and then after that we can just scale it up by 4 in the vertical direction after.

There is another thing we should notice. Consider a quadratic such as $x^2 - 7x + 10$, in which if we consider the equation $x^2 - 7x + 10 = 0$ this has roots/solutions (2 and 5 – can be found by factoring or using the formula, if you look for numbers that add to 7 and multiply to 10 you will see that 2 and 5 work). What this means is that if we sketch the graph $y = x^2 - 7x + 10$ it should cross through (2, 0) and (5, 0). Indeed this is exactly what happens, see figure 25.


Figure 25

Now we can solve inequalities with terms like x^2. Let's do an example.

Lets find all x such that $x^2 + 11x + 6 < 4x$. First, lets move everything to one side so that we have 0 on one side. This gives $x^2 + 7x + 6 < 0$. Now lets solve $x^2 + 7x + 6 = 0$. We can factor it – looking for 2 numbers that add to -7 and multiply to 6 we see that -1 and -6 work so our factorization is – being careful to get the minus signs all straight – is $(x - (-1))(x - (-6)) = (x+1)(x+6)$. Therefore the solutions are $x = -1$ and $x = -6$. By previous work, we know that the graph of this will look like a parabola, and it will cross the x-axis exactly at the points -1 and -6. Figure 26 shows the graph of this.


Figure 26

We see that -1 and -6 are where this crosses the x-axis. Therefore, the solution to $x^2 + 7x + 6 < 0$ is exactly when $-6 < x < -1$. Note that the inequality is $x^2 + 7x + 6 < 0$ and not $x^2 + 7x + 6 \leq 0$, so the points -1 and -6 are not solutions, and the solution is not $-6 \leq x \leq -1$. Read test questions carefully as you don't want to get caught out by this. One time I failed a test because I read the inequality the wrong way around, I was so mad.

18 Series

Lets do a simpler proof involving algebra.

Proposition. If x is an integer then x^2+x is even.

Proof. We factor x^2+x as x(x+1). If x is even this is an even number times an integer which is even. If x is odd then this is an integer x times x+1 which is even and thus even.

□

Note that an equivalent statement to the proposition above is that $\frac{n^2+n}{2}$ is an integer for every integer n, however we will now show this in a more elegant way. For the first few integer values of n, $\frac{n^2+n}{2}$ gives the sequence 1, 3, 6, 10, 15, 21, 28, ... But notice this is just 1, 1+2, 1+2+3, 1+2+3+4, ... So lets prove that this pattern always holds.

There is a nice picture that proves this result.

Proposition. $1 + 2 + 3 + ... + n = \frac{n^2+n}{2}$

Proof.

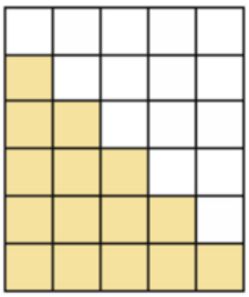


Figure 27

In figure 27 there are 1+2+3+4+5 yellow squares, in general we could make an analogous image for n more than 5. But notice that the number of yellow squares is half the total number of squares (it is easy to see this), but that is the number of rows (6, or in general n+1), times the number of columns (5, or in general n). Therefore, 1+2+3+...+n is the number of yellow squares which is half of the total number of squares which is half of $n(n + 1)$ so we get the formula

$$1 + 2 + 3 + ... + n = \frac{n^2 + n}{2}$$

□

As an example, the sum of every number up to 100 is $\frac{100(100+1)}{2} = \frac{100*101}{2} = \frac{10100}{2} = 5050$ (gauss reference – legend says that guy figured it out instantly with a similar argument when the teacher expected him to take a long time).

A sequence of the type where each term is some number more than the previous one (like the 1, 5, 9, 13, ... example) is called an arithmetic sequence. There is a formula for the sum of the first n terms in an arithmetic sequence, but first we need some notation.

I can write, for example

$$\sum_{n=1}^{100} 4n - 3$$

The symbol on the left means "Sum", $n = 1, 100$ on the bottom and top means "Where n ranges from 1 to 100", so what

we need to do is add $4n-3$ as n goes from 1 to 100. Ie, we have to calculate:

$$4(1)-3+4(2)-3+4(3)-3+\cdots+4(99)-3+4(100)-3$$

Or

$$1+5+9+13+\cdots+393+397$$

If we start with a sum like the one above, we need to find an expression for the n'th term as discussed earlier, then determine the number of terms, in this case the last term is 397 and the expression is (from earlier) $4n-3$, so the last term is the n'th term where $397=4n-3$, so by simple algebra $n=100$.

We can split the sum as follows:

$$\sum_{n=1}^{100} 4n-3=\sum_{n=1}^{100} 4n-\sum_{n=1}^{100} 3=4\sum_{n=1}^{100} n-\sum_{n=1}^{100} 3$$

Now the sum on the right is just adding 3 100 times as it is independent of n, so it is just 300. And the sum of the left is 5050 from earlier, or $\frac{100^2+100}{2}$ by that formula we derived. So

$$1+5+9+13+\cdots+393+397=\sum_{n=1}^{100} 4n-3=4*5050-300=19900$$

Note that the variable we are summing over, in this case n, will never appear in the final expression for the sum: It is a "dummy" variable that we introduce just for this purpose. What I mean by this will become more clear by noticing the following:

$$\sum_{n=1}^{100} 4n-3=\sum_{r=1}^{100} 4r-3=\sum_{k=1}^{100} 4k-3=\sum_{\text{skibiditoilet}=1}^{100} 4\text{skibiditoilet}-3$$

Don't question the last index, I need to cater to the children.

The point is, the index I am summing over should not come up in the final answer because it does not matter what I name it, as the expressions above all mean the same thing.

In general

$$\sum_{n=1}^{K} An+B=A\sum_{n=1}^{K} n+\sum_{n=1}^{K} B=A\left(\frac{k^2+k}{2}\right)+Bk=k\left(A\left(\frac{k+1}{2}\right)+B\right)$$

Where $\frac{k^2+k}{2}$ comes from the formula we derived earlier.

Note that the first term (a) is A+B and the common difference between terms (d) is A, so the formula can be rewritten as $\frac{n}{2}[2a+(n-1)d]$ by rearranging where n is the number of terms, d is the difference between the terms, and a is the first term.

Also, we can define factorials as follows (by putting an exclamation mark after a number, or saying "number factorial": When someone puts an exclamation mark after a number for emphasis you will never see it the same way after this):

1!=1

2!=1*2=2

3!=1*2*3=6

4!=1*2*3*4=24

5!=1*2*3*4*5=120

And so on.

19 More geometry

19.1 Sine rule

Lets suppose we have a triangle with side lengths a, b, c and angles A, B, C opposite those sides respectively, as shown in Figure 28.

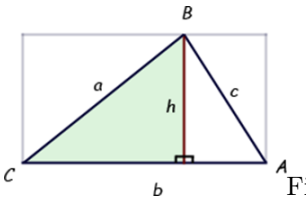


Figure 28

Theorem. We have the following tools which we can use to solve geometrical problems

1. The area of the triangle is $\frac{1}{2}ab * \sin(C)$
2. $\frac{\sin(A)}{a} = \frac{\sin(B)}{b} = \frac{\sin(B)}{c}$

Proof. the area is given by $\frac{1}{2}bh$ from earlier, but notice: $\sin(C) = \frac{h}{a}$ so $h = a\sin(C)$, so the area of the triangle is $\frac{1}{2}ab * \sin(C)$. So, if we know an angle and the two side lengths adjacent to that angle, we can work out the area of the triangle.

Note that the same argument would work even if the h line coming from B does not end up on the line segment AC.

Now notice that if we rotate the triangle and apply the same argument, we will get that the area is equal to $\frac{1}{2}ca * \sin(B)$ and $\frac{1}{2}bc * \sin(A)$. Now these are all equal to the area of the triangle, hence equal to eachother, hence by doing some algebra there is a little rule we can derive:

$$\frac{1}{2}ca * \sin(B) = \frac{1}{2}ab * \sin(C)$$

Since a is not 0 we can divide everything by $\frac{a}{2}$, so

$$b * \sin(C) = c * \sin(B)$$

Now we will divide by $b * c$ since it is non-zero to get

$$\frac{\sin(C)}{c} = \frac{\sin(B)}{b}$$

But by exactly the same argument with a and b instead of c and b, we get the following triple equality:

$$\frac{\sin(A)}{a} = \frac{\sin(B)}{b} = \frac{\sin(C)}{c}$$

as claimed.

□

This is called the sine rule. It is important because if we know any angle and the length of the side opposite it, then once we have another angle or side we can work out its corresponding angle or side by solving for it using this rule. However, note that knowing the sin of an angle does not uniquely determine it, for example $\frac{1}{2} = \sin(30°) = \sin(150°)$. This is for 2 reasons:

The graph of $y = \sin(x)$ is symmetric about the line $x = 90°$

If we walk 30 degrees around a circle we will have the same y coordinate as when we have 30 degrees left to reach 180 degrees around the circle. This is actually the reason that the reason above is true.

Note that for sin, cos and tan we sometimes write $\sin(\theta)$ instead of $\sin(x)$ because θ represents angles.

Therefore when applying the sine rule, we may have enough information to narrow down two possible triangles but not any further, and extra information may be given in the context of the question to allow unique determination of the triangle.

19.2 Cosine rule

Note that if x goes from 0 to 180 degrees then cos(x) goes from -1 to 1 so knowing the cosine does uniquely determine the angle. This is good because it means that for the cosine rule I am about to introduce, we do not have the ambiguous case.

Theorem. If we are in the setup above, then

$$c^2 = b^2 + a^2 - 2ab * \cos(C)$$

Note that if C is 90 degrees this is just pythagoras. This is called the cosine rule.

Proof.

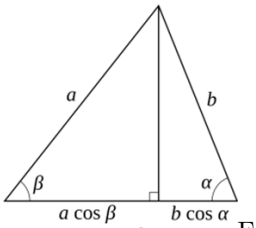


Figure 29

We draw the altitude of the triangle perpendicular to c through the opposite vertex. From figure 29 we see that

$$c = a * \cos(\beta) + b * \cos(\alpha)$$

Multiplying by c,

$$c^2 = ac * \cos(\beta) + bc * \cos(\alpha) \tag{1}$$

We can consider either a or b as the base to get

$$a^2 = ac * \cos(\beta) + ab * \cos(\gamma) \tag{2}$$

$$b^2 = bc * \cos(\alpha) + ab * \cos(\gamma) \tag{3}$$

Note that γ is what I'm calling the angle of the triangle that is not α or β.

We want to show that

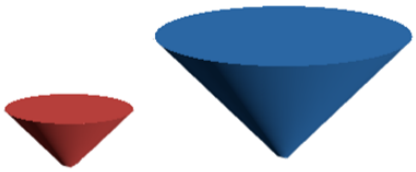
$$c^2 - (b^2 + a^2) = -2ab * \cos(C)$$

But this follows from what we derived above and some algebra. Specifically, by subtracting equations (2) and (3) from equation (1).

□

19.3 Scaling shapes

Now suppose we have a shape and then we scale it by a factor of X. Figure 30 is an image to show exactly what I am talking about, the blue cone is twice as tall and wide as the red one. We want to consider how much the volume of the blue cone compares to the volume of the red cone.


Figure 30

Well, we could first stretch it by 2* in the left-right direction, this multiplies the volume by 2, then we could do that in the front-back direction, now the volume is 4* bigger, then we could do that in the up-down direction, now the volume is 8* bigger. By this principle, if we scale something by a factor of X and we have n dimensions, the area/volume is multiplied by X^n. Also, surface area is like a dimension 2 object so it is multiplied by X^2. However, defining surface area rigorously and proving this for all cases is something very difficult that we will do in undergraduate levels. Therefore, we will show this for only a few shapes (For shapes with flat faces it's immediate, for spheres and cones it comes from the upcoming formulae. On school tests you can usually assume you are being asked about one of these shapes or combinations of them)

19.4 Formulae for volumes and surface areas

Now, informally, the surface area of a shape is the area of its exterior, or how much paint you would need to cover it. What this means is that my cat has an unfathomably large surface area because she is very soft. Anyway, we can calculate the surface area of some simple shapes:



Figure 31

This cuboid shown has height 2, depth 3 and length 5. The surface area of this is the sum of the areas of all its faces:

$$2 * 3 + 2 * 3 + 2 * 5 + 2 * 5 + 3 * 5 + 3 * 5 = 62.$$

Now there are some formulas for surface areas and volumes of shapes – the proofs for these are all deferred to levels 3 and 4.

Theorem. The following are true:

1. The volume of a sphere with radius r is $\frac{4}{3}\pi r^3$
2. The surface area of a sphere with radius r is $4\pi r^2$
3. The volume of a cone with height h and base circle having radius r is $\frac{1}{3}\pi r^2 h$
4. The surface area of a cone with height h and base circle having radius r is $\pi r^2 + \pi rl$, πr^2 for the base from the area of a circle and πrl for the curved part. Here l means the distance if you measure from the base to the tip of the cone.

Remark. We know that areas will be proportional to r^2 and volumes proportional to r^2, so the only hard part to prove is what the exact constants are that we multiply by.

Now something that I can actually prove here:

Proposition. The volume of a cylinder is the area of the base times the height, and the surface area of the curved portion of a cylinder is the circumference of the base times the height, ie we get the following:

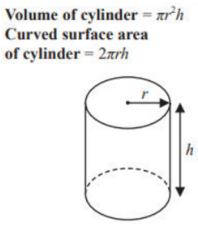


Figure 32

Proof. Immediate from figure 32.

□

Note that volume and surface area have fundamentally different units – We are taking about units of area (squares) and units of volume (cubes). It does not make sense to add an area to a volume for this reason.

These formulas can be used to

i) Calculate the surface area of a shape

ii) Calculate the radius or height of a shape given the surface area and other relevant information.

19.5 Circle properties

These properties are useful for solving for lengths and angles.

Note that if I have 3 points in 2D space and they do not lie on the same line, it is easy to see that there is a unique circle whose circumference goes through those points. We will expand on this idea later.

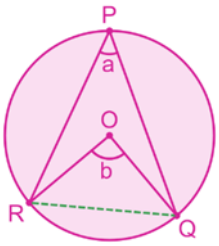


Figure 33

Figure 33 shows a circle with center O. A very important theorem (which we will prove, then use it to derive the rest of our circle properties) says that in a configuration like this, the angle b is double the angle a.

The theorem also works even if we have a situation like figure 34 or figure 35.

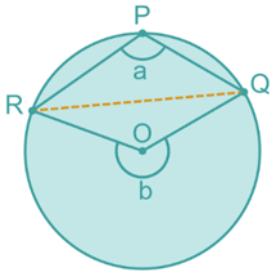


Figure 34

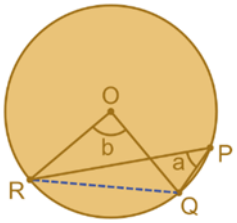


Figure 35

Theorem. If we have a situation like figure 33, 34 or 35, and the angles are named accordingly, then b=2a.

Proof. In order to do this we need a way of naming angles. Note that in these diagrams the angle a is the angle between the line RP and the line PQ, and P is the point that the angle is at. We write the angle as RPQ. The point the angle is at is the second letter, and the first and third letter are other points on the lines that the angle is between.

Now in figure 33, draw a line from O to R. Then OP, OR and OQ (note that OP means the distance from O to P and similar for the others) are all equal to the radius of the circle, so their lengths are equal to eachother. Therefore the triangles OPR and OPQ (meaning the triangle you get by connecting those points in order) have two sides of the same length. This type of triangle is called an isoseles triangle. Note that it is now easy to see that the triangle OPR is symmetrical in the sense that it will not change if we reflect it about the line through O perpendicular to PR, and thus the angles ORP and OPR are the same. This is a general property of isoseles triangles – two of their angles are the same. Therefore, since the angles of the triangle OPR (meaning the triangle with vertices O, P and R) must add up to 180 degrees, it means that $2OPR + POR = 180°$. By the same argument, $2OPQ + POQ = 180°$. We can add these equations together to get $2OPR + 2OPQ + POR + POQ = 360°$. But we know that the angles b, POQ and POR form a circle and thus $b + POQ + POR = 360°$. But we also know that $OPR + OPQ = a$, and thus we have that $2a + 360° - b = 360°$ by putting all the last 3 equations we got together. Simplifying this gives $2a = b$ as required, and the same argument works for figure 34. Thus we just need to deal with figure 35.

For that, note that $OPQ - OPR = a$, so we want to show that $2OPQ - 2OPR = b$. Lets call E the point where the line segments RP and OQ intersect, then $a + OQP + PEQ$ is 180 degrees since they are the angles of a triangle. It is easy to see from the diagram that the angles PEQ and OER are equal. Also since $a + OQP + PEQ = 180°$, $PEQ = 180° - a - QOP$ so we know that $OER = 180° - a - QOP$. But we know that $b + OER + ORP = 180°$ since those are the angles in the triangle OER. Hence, by the equation for OER, $b + 180° - a - QOP + ORP = 180°$, but from the fact that $OPQ - OPR = a$, we know that $OPQ - ORP = a$ as $OPR = ORP$ by symmetry of the isoseles triangle ORP – same argument as above. Hence, $b + 180° - a - QOP + ORP = 180°$ which means $b + 180° - 2a = 180°$ and therefore finally we get $b = 2a$.

□

What we have done above is called angle chasing.

Now, as promised, from this theorem, we will see that the rest of the properties we will show come out easily and we have done the hard work.

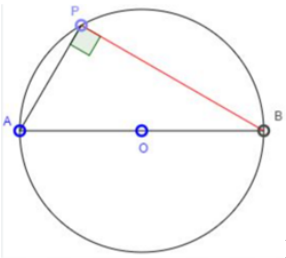


Figure 36

Figure 36 demonstrates the next circle property we will prove.

Theorem. If we have a triangle formed by three points that determine a circle and it just so happens that one of the sides of the triangle is a diameter (ie, passes through the center of the circle), then the triangle is right angled (ie, has a 90 degree angle).

Proof. With reference to figure 36 and the previous theorem, we know that $180° = AOB = 2APB$ since angles on a straight line are 180 degrees, hence $APB = 90°$.

□

Note also that if we have a right angled triangle and we put a circle that goes through its three points, then its hypotenuse (long side) will be the diameter of the circle. This is because (using the same point names as in the reference image) if AB was not the diameter, then the angle AOB would not be 180 degrees so the angle APB would not be a right angle.

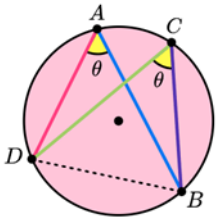


Figure 37

Figure 37 shows another circle property, which is that angles coming from the same segment are equal. This is because if we call the center O, then by the first theorem, the angle DOB is both twice DAB and DCB using the point names from the reference image, hence DAB and DCB are equal to eachother.

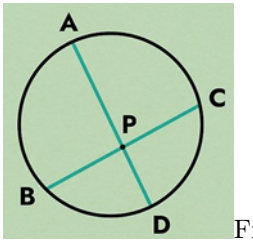


Figure 38

Figure 38 shows another circle property.

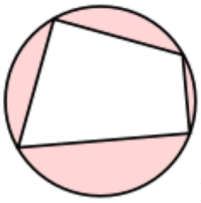
If we have two intersecting chords of a circle, as in figure 38, then

$$(AP)(DP) = (BP)(CP)$$

Proof. Referencing figure 38 to name things, if we make APB and PCD into triangles then notice that by the previous theorem, ABC and ADC are the same, and from the diagram APB and CPD are the same, and therefore CPD and APB are similar triangles since they share 2 angles – ie you can get from one to the other by reflecting and scaling it. Therefore, the scale factor between the two triangles is exactly $\frac{PD}{PB}$, but also $\frac{PC}{PA}$, therefore $\frac{PD}{PB} = \frac{PC}{PA}$. Multiplying both sides by $(AP)(BP)$ gives exactly the desired result.

□

Now we will return to the "A circle is determined by 3 points" idea and talk about when it is determined by 4 points. So suppose we have 4 points that determine a circle. Now draw a quadrilateral with those 4 points as vertices, as shown in figure 39:


Figure 39

A quadrilateral whose vertices determine a circle is called a cyclic quadrilateral. We know its angles add up to 360 degrees but we can say something stronger: Suppose the vertices are A, B, C, D in clockwise order and the center is O. Then I will name some angles in figure 40:

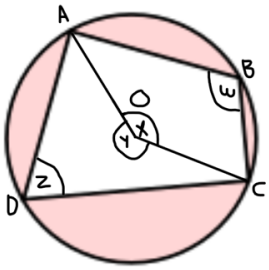


Figure 40

From figure 39, we see that $x + y = 360$, and from the "$b = 2a$" theorem above, $x = 2z$ and $y = 2w$, hence $z + w = 180$. For this reason, any two opposite angles in a cyclic quadrilateral add up to 180 degrees.

Proposition. Suppose a quadrilateral has the property that each opposite angle pair adds up to 180 degrees. Then it is cyclic.

Proof. Consider taking a quadrilateral $ABCD$ with this property and putting a circle through three of the points A, B, C which we know we can do for sure. We haven't technically proven this, but it is easy to see. Then we want to show that D is on the circle. Let D' be the point where AD intersects the circle (Of course, $D = D'$ but we don't know this as it is what we want to prove, this is a common trick in geometry). Then $ABCD'$ is a cyclic quadrilateral, but then $ABC + AD'C = 180°$, and since we assume $ABC + ADC = 180°$ it means $ADC = AD'C$. But then because $ADC = AD'C$ and A, D and D' are parallel and D' and D are on the same side of A on the line AD, it is easy to see geometrically that $D'C$ is parallel to DC if the angles satisfy $ADC = AD'C$, but if $D \neq D'$ this only happens if D' lies on the line DC, but since it also lies on the line AD by assumption, D' is the unique intersection of the lines AD and DC which is D.

□

We have one last circle theorem called the alternate segment theorem.

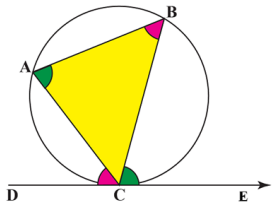


Figure 41

Theorem. If we draw a line tangent to a circle (ie, perpendicular to the radius at the point of contact, and for there to only be a single point of contact where it barely touches the circle) then mark angles as in figure 41, then $CAB = BCE$ and $ABC = ACD$.

Proof. We will just prove that $ACD = ABC$ as the other equality follows from the fact that $ACD + ACB + BCE$ is 180 degrees and so is $ABC + ACB + BAC$ as those are the angles of a triangle, and thus we know that in fact $ACD + ACB + BCE = ABC + ACB + BAC$ and thus $ACD + BCE = ABC + BAC$, so if we can prove $ABC = ACD$ then it will follow that also $BAC = BCE$.

By a theorem above, we can move B around on the circle and the angle ABC will not change. So lets make the convenient choice to move B such that BC is perpendicular to the line DE. If the line AC were in the way, we would simply move that one instead and do the same argument. Now once we have done that, BAC is a right angle as BC is the diameter of the circle and we can apply the relevant theorem that we proved earlier. But DCB will also be a right angle. We have $ABC + BAC + ACB = 180°$ so $ABC + ACB = 90°$, but also $ACB + ACD = 90°$, so finally we get that $ABC + ACB = ACB + ACD$ and hence $ABC = ACD$ which is what we wanted to show.

□

Those are all the circle properties we cover in this level.

20 Fractions and decimal expansions

You may or may not have realized, or know, that if you take any fraction and write out its decimal expansion, ie write out the number, it will settle into a perfect repeating pattern. Lets prove this. Although this proof is not required for GCSE maths or equivalent, we provide it here.

Theorem. If a number can be written as a fraction of two integers then its decimals eventually repeat. Also, if a number's decimals eventually repeat, then it can be written as a fraction of two integers.

Proof. We will prove the first part by demonstrating why this is always true using an example. The reason we talk about all this is because given a repeating decimal, we want to be able to convert between recurring decimals and fractions and it is very important that we know that this is always possible.

Consider $\frac{1}{7} = 0.142857142857142857\ldots$. Then $\frac{10}{7} = 1.42857142857142857\ldots$. This does not start with a 0 so what we do is we subtract 1 until it does, and each time we do this we need to subtract 7 from the numerator to keep the equation balanced. We get $\frac{3}{7} = 0.42857142857142857\ldots$. We can repeat this procedure again: $\frac{30}{7} = 4.2857142857142857\ldots$. so $\frac{2}{7} = 0.2857142857142857\ldots$. Each time, the numerator will be either 0, 1, 2, 3, 4, 5, or 6. At some point, it must go back to somewhere it was previously since there are only finitely many possible remainders, and this will mean it must go into an endless loop as each numerator is uniquely determined by the previous one and only the previous one, and hence the decimal places will also go into an endless loop.

Also, even if the numerator did not start at 1, I could have subtracted 7 until it was strictly less than 7 and applied the same argument. I could have also applied the same argument if the denominator was any other non-zero integer to begin with.

Note that if a number's decimals do not repeat, it therefore means the number is irrational: it is not a fraction of integers. $1.01001000100001\ldots$ is irrational as it does not technically repeat. So is pi, which we prove in the misc results section of this website, hence pi's decimals do not start to infinitely repeat.

We know that long division lets us convert a fraction into a recurring decimal, but we can do the opposite. To do this, we will – much to the dismay of u/SouthPark_Piano if you happen to get this reference – use the fact that $0.999999999\ldots = 1$, as we proved in level 1, or alternatively because they would have to occupy the same position on the number line.

Because of this fact, we see that, eg $0.22222222222\ldots = \frac{2}{9}$. Also, $0.259259259259\ldots = \frac{259}{999}$. However, it just so happens that $\frac{259}{999}$ can be simplified to $\frac{7}{27}$ (Factor 259 and 999 into primes). Also, even if a decimal number does not start with 0 or only eventually repeats, we can still do this. Here is an example that addresses both:

$$4.478181818181\ldots = 4 + 0.47 + 0.0081818181\ldots = 4 + 0.47 + \frac{1}{100} * 0.8181818181\ldots = 4 + \frac{47}{100} + \frac{1}{100} * \frac{81}{99}$$

Now we just add and multiply and simplify the fractions as we should know how to do and we get that the correct answer is $\frac{2463}{550}$.

From this procedure it is easy to see the second part of the theorem above – that any eventually repeating decimal number can be written as a fraction.

□

21 More algebra

21.1 Difference of squares rule

It is useful to know the difference of squares rule: If you have the difference of two squares you can factor it. Ie, by simple algebra you can verify that $a^2 - b^2 = (a+b)(a-b)$ for any a and b. Therefore in an expression like $4x^2 - 9$ you can factor it as follows: Spot that $4x^2$ is the square of $2x$ (Not the square of $4x$, as $(4x)^2 \neq 4(x^2)$) and 9 is the square of 3, so we get $4x^2 - 9 = (2x-3)(2x+3)$ by applying this rule.

21.2 Algebraic fractions

It is possible to add, subtract, multiply and divide and simplify algebraic fractions the same way you do with normal fractions. Here are some quick examples of this:

$\frac{x^2-5x+6}{x^2-9}$ can be simplified. To do this we will factorize the numerator (top) and the denominator (bottom). If we do that we get $\frac{(x-2)(x-3)}{(x+3)(x-3)}$. Therefore, for all x not equal to 3 or -3, this is just $\frac{x-2}{x+3}$.

Example. Lets solve the intimidating looking equation $\frac{x+6}{x+1} - \frac{1}{2x-4} = 4$ (x is not -1 or 2 so we are not dividing by 0). As with fractions that just involve numbers, we want to get a common denominator. We can do this by first multiplying everything by $x+1$, then multiplying everything by $2x-4$.

Multiplying by $x+1$: $x + 6 - \frac{x+1}{2x-4} = 4(x+1)$

Multiplying by $2x-4$: $(x+6)(2x-4) - (x+1) = 4(x+1)(2x-4)$

Expanding: $2x^2 + 7x - 25 = 8x^2 - 8x - 16$. Moving everything to one side gives $6x^2 - 15x + 9 = 0$. Everything there is a multiple of 3 so we can simplify to $2x^2 - 5x + 3 = 0$. Solving this by either the quadratic formula, or spotting the factorization $(2x-3)(x-1)$, gives the solution: $x = 1$ or $x = \frac{3}{2}$.

21.3 Simultaneous equations

We can solve simultaneous equations that involve quadratics.

Example. $y = 11x - 2$ and $y = 5x^2$. Geometrically, we want to find where a line and parabola intersect, and we want to find x and y. From these equations, $5x^2 = 11x - 2$. To apply the quadratic formula we want to get 0 on the right hand side so we must move everything over: $5x^2 - 11x + 2 = 0$. The solutions in x to this are $x = \frac{1}{5}, x = 2$. For each of these values, we can use $y = 11x - 2$ to get that for $x = \frac{1}{5}, y = \frac{1}{5}$ and for $x = 2, y = 20$.

$y = 2x - 11$ and $x^2 + y^2 = 25$. Luckily we know what y is in terms of x from the first equation so we can put it directly into the second equation: $x^2 + (2x-11)^2 = 25$ which we can expand to get $5x^2 - 44x + 121 = 25$ so $5x^2 - 44x + 96 = 0$. I figure I should actually give an example of how the quadratic formula works in practice, so here is that example:

$a = 5, b = -44, c = 96$, so

$x = \frac{-(-44) \pm \sqrt{(-44)^2 - 4*5*96}}{2*5} = \frac{44 \pm \sqrt{16}}{10} = \frac{44 \pm 4}{10}$ which is 4 or $\frac{48}{10} = \frac{24}{5}$.

For each of these values of x, we can use $y = 2x - 11$ to work out y.

22 Vectors

A vector is basically an arrow in space. These will seem weird now but will be useful later.

For the purposes of this level you just need to know that a vector can be thought of as an arrow in 2 or 3 or higher dimensional space that starts from the origin or the point $(0,0)$ and ends at some other point. You could also move the arrow to start somewhere else but it would be the same vector - you can think of adding vectors by putting the tail of a second arrow at the tip of the first one and looking at where the tip of the second one ends up.

We typically write vectors as columns with each entry as the coordinate, or component. Precisely what I mean is, for example, if a vector in 2D space goes from $(0,0)$ to $(2,1)$ then we write it as $\begin{pmatrix} 2 \\ 1 \end{pmatrix}$.

We can add and subtract them by adding and subtracting each of the components and we can multiply them by a fixed number. Adding two vectors is basically putting the tail of one at the tip of another and finding where the tip of the second one is.

Example: In 2D the equation of a vector that goes from $(2,1)$ to $(5,-2)$ would be the vector that you need to add to $\begin{pmatrix} 2 \\ 1 \end{pmatrix}$

to get to $\begin{pmatrix} 5 \\ -2 \end{pmatrix}$ which is $\begin{pmatrix} 5 \\ -2 \end{pmatrix} - \begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ -3 \end{pmatrix}$. If A is the point $(2, 1)$ and B is the point $(5, 2)$ then we would write this as $\overrightarrow{AB}$.

Also, if we have a translation where we move an object in space, such as a graph, we can use a vector to describe what we move it by.

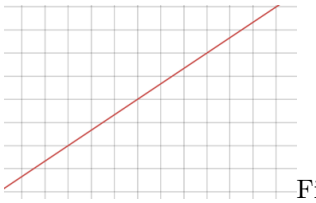
Vectors have length in the usual sense.

Example: $\begin{pmatrix} 9 \\ 24 \\ 32 \end{pmatrix}$ by pythagoras has length $\sqrt{9^2 + 24^2 + 32^2} = \sqrt{1681} = 41$

In theory you can solve geometrical problems using vectors.

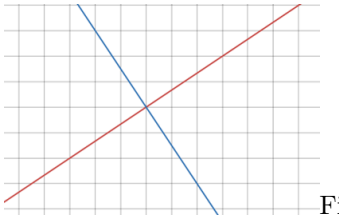
23 Slopes and perpendicular lines on graphs

Example. To find the slope of the line in figure 42 which is assumed to be to scale, we note that in order for the line to go up 2 units it must go forward 3 units, and the slope is "rise over run" so we get that the slope is $\frac{2}{3}$


Figure 42

Now try to think about the line that would make an angle of 90 degrees with this line and try to guess what the slope will be before reading on.

I will now add a perpendicular line to the plot, see figure 43.


Figure 43

The slope of the blue line is $-\frac{3}{2}$ by the same reason as above.

The general principle which holds for precisely the same geometric reason is the following key fact:

Proposition. If a line has a slope of m, then the line perpendicular to that line has a slope of $-\frac{1}{m}$ (Unless $m = 0$ because you can't divide by 0)